Problem 1

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Problem 1 *For the following problems, the derivative is given. Determine which function was the original function.*

(a) *The derivative is* $f'(x) = \cos(x)e^{\sin(x)}$*. Which is the original function?*

(i) $f(x) = (\sin(x))(e^x)$

(ii) $f(x) = \sin(e^x)$

(iii) $f(x) = e^{\sin(x)}$

(iv) $f(x) = e^{x\sin(x)}$

Explanation. Since

$$\frac{d}{dx}\left(e^{\sin(x)}\right) = e^{\sin(x)} \cdot \cos(x) = \cos(x)e^{\sin(x)}$$

the correct answer is (iii).

(b) *The derivative is* $g'(x) = 4\left(\tan(x^4 - 5x)\right)^3 \sec^2(x^4 - 5x)(4x^3 - 5)$*. Which is the original function?*

(i) $g(x) = (\tan(x) - 5x)^4$

(ii) $g(x) = \tan^4(x) - 5x^4$

(iii) $g(x) = \tan(x^4 - 5x)$

(iv) $g(x) = \tan^4(x^4 - 5x)$

Explanation. Since

$$\frac{d}{dx}\left(\tan^4(x^4 - 5x)\right) = 4\tan^3(x^4 - 5x)\sec^2(x^4 - 5x)(4x^3 - 5)$$

the correct answer is (iv). Notice that $\tan^3(x^4-5x) = \left(\tan(x^4 - 5x)\right)^3$ are (slightly) different notations for the exact same expression.